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Optimizing short-acting β 2-Agonist therapy in acute asthma exacerbations: A comprehensive analysis of clinical predictors

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ABSTRACT

Background and Aim: Asthma is a chronic respiratory disease marked by airway inflammation and narrowing, leading to acute exacerbations. Effective management, including the use of short-acting β 2-agonists (SABA), is crucial for symptom control. Factors like inhaled corticosteroid (ICS) adherence, exacerbation severity, and respiratory infections may influence SABA effectiveness. This study explores these clinical predictors in managing acute asthma exacerbations.

Materials and Methods: A cross-sectional study was conducted at Wahidin Sudirohusodo General Hospital and its network of affiliated healthcare centres in Makassar, Indonesia, with 103 adult asthma patients. Demographic, clinical data, asthma control status, peak expiratory flow (PEF), and oxygen saturation (SpO_2) before and after SABA administration were collected. ICS adherence, exacerbation severity, and respiratory infection status were also evaluated. Paired t-tests and chi-square tests were used for statistical analysis.

Results: Significant improvements in SpO_2 (from 91.2% to 97.0%) and PEF (from 250.7 L/min to 331.6 L/min) were observed after SABA treatment. ICS adherence was a significant factor influencing treatment outcomes, with adherent patients showing better responses. Respiratory infections negatively impacted exacerbation resolution. Exacerbation severity and obesity lost significance after adjusting for ICS adherence and infections.



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Discussion: ICS adherence was the primary factor influencing SABA response. Respiratory infections worsened outcomes by increasing airway inflammation, reducing bronchodilator efficacy. Exacerbation severity and obesity were less predictive when adjusted for ICS adherence and infection status.

Conclusion: Optimizing ICS adherence and controlling respiratory infections are essential to improving SABA effectiveness in acute asthma exacerbations. A comprehensive strategy addressing these factors will enhance treatment outcomes. (www.actabiomedica.it)

Key words: asthma exacerbation, inhaled corticosteroids (ICS), bronchodilator response, medication adherence, respiratory infections

Introduction

Asthma is a chronic respiratory condition characterized by inflammation and narrowing of the airways, leading to airflow obstruction. This condition not only significantly impairs the quality of life but also poses life-threatening risks if not effectively managed. While asthma remains incurable, appropriate management can control symptoms and prevent exacerbations, allowing individuals to lead relatively normal lives (1,2). The symptoms commonly include shortness of breath, wheezing, coughing, and a feeling of chest tightness, which tend to worsen during the night or upon waking. These symptoms are triggered by a variety of factors, including allergens, physical exertion, and viral respiratory infections (3). Asthma is a global health issue, affecting over 300 million people worldwide, with prevalence rates ranging from 1% to 18% across different populations. In Indonesia, approximately 2.4% of the population is diagnosed with asthma, a condition that frequently leads to emergency room visits due to acute exacerbations and respiratory complications (3). Early detection of exacerbations is critical to preventing severe outcomes, and tools such as peak flow meters play a vital role in this process by measuring peak expiratory flow rate (PEFR). A decline in PEFR values can serve as an early indicator of worsening asthma, allowing for timely intervention. The management of asthma often involves the use of bronchodilators,

with short-acting β_2 -agonists (SABA) like salbutamol being the cornerstone of acute asthma treatment. However, the overuse of SABAs, especially as a sole treatment, raises concerns about inadequate asthma control and potential misuse. In response, recent Global Initiative for Asthma guidelines now recommend combining SABA with inhaled corticosteroids (ICS) to provide more comprehensive and sustained symptom management (2,4). Despite the shift toward ICS use, nonadherence to controller therapies remains a significant barrier to effective asthma management. Lack of adherence to ICS regimens is closely associated with increased frequency of asthma exacerbations and suboptimal symptom control, underlining the need for better patient education and support (5,6). During severe exacerbations, the responsiveness to bronchodilators such as SABA can be influenced by various clinical factors. For instance, low oxygen saturation ($\text{SpO}_2 < 92\%$) at presentation has been shown to predict a poor response to SABA treatment, while a history of prior intubation for asthma exacerbations also correlates with treatment failure (7). Moreover, factors such as obesity, age, and gender play critical roles in the severity of asthma and its response to bronchodilator therapy. Specifically, obesity has been linked to a higher incidence of asthma and more frequent exacerbations, making it a crucial factor to consider in asthma management (8,9). Additionally, asthma in the elderly is associated with more severe outcomes, including increased

morbidity and mortality, highlighting the need for age-specific approaches to treatment (10,11). This study aims to investigate the factors that influence the effectiveness of SABA in managing acute asthma exacerbations. By analyzing clinical parameters such as oxygen saturation (Sp_2O), peak expiratory flow (PEF), and the resolution status of exacerbations, this research seeks to deepen our understanding of the variables that impact bronchodilator response, with the goal of improving asthma management and patient outcomes.

Materials and methods

This study was conducted to assess the factors influencing the effectiveness of SABA in treating acute asthma exacerbations. The research was carried out across multiple healthcare centres, including Wahidin Sudirohusodo General Hospital, a tertiary referral hospital in East Indonesia, and its network of affiliated healthcare centres in Makassar, Indonesia. Ethical

Study design and population

This was a multi-centre, cross-sectional study involving adult patients aged 18 years and older who were diagnosed with asthma according to the criteria outlined in the 2019 Guidelines for Diagnosis and Management of Asthma in Indonesia. The guidelines specify that asthma should be diagnosed based on the presence of symptoms such as wheezing, coughing, shortness of breath, and chest tightness, along with a history of variable airflow obstruction. A confirmed diagnosis is supported by objective measurements, such as spirometry or peak flow measurements, which show reversible airflow limitation. Patients must also demonstrate a positive response to bronchodilators (1,2).

Inclusion criteria

Adult patients (18 years or older) diagnosed with asthma according to the 2019 Indonesian Asthma Guidelines, who had experienced an acute exacerbation within the past 24 hours, characterized by worsening symptoms (wheezing, shortness of breath, or cough) and the need for SABA treatment.

Exclusion criteria

Patients who were current or former smokers were excluded from the study. Only never smokers were included. Additionally, patients with a history of bronchiectasis, confirmed by clinical assessment and chest CT scans, were excluded due to the confounding effect of bronchiectasis on asthma exacerbations. Patients with a history of atrial fibrillation, identified through medical history and ECG findings, were also excluded due to the potential interference of this condition with asthma management. Furthermore, patients with any contraindications to SABA therapy (such as hypersensitivity or severe adverse reactions to beta-agonists) were excluded, based on medical records and clinical evaluation. Patients with significant comorbidities that could affect the study outcomes, including chronic obstructive pulmonary disease (COPD), heart failure, or active respiratory infections other than asthma, were also excluded.

Data collection

Demographic and clinical data, including age, gender, body mass index (BMI), and asthma control status, were recorded. The asthma control status was assessed using the Asthma Control Test (ACT), a validated measure of asthma control (12).

Measurement of asthma exacerbation

Asthma exacerbation was assessed based on clinical criteria: the use of SABA as a rescue medication due to worsening symptoms, PEF measurements, and oxygen saturation (SpO_2). PEF was measured using a ROSSMAX® Peak Flow Meter, which provides a quick measure of the maximum speed of exhalation. A drop in PEF from the patient's personal best value indicated a worsening of asthma control. Oxygen saturation was measured using a pulse oximeter (Masimo® Rad-5 Pulse Oximeter, Masimo Corporation), and a SpO_2 value of less than 92% was considered indicative of a severe exacerbation, requiring urgent intervention. Viral infections were assessed using throat swabs (COPAN Eswab), following standard clinical procedures. The swabs were used to detect common viral pathogens,

including rhinovirus, influenza, and respiratory syncytial virus (RSV), which are known to exacerbate asthma. This method was chosen for its ability to identify these viral agents and, in addition, provide indirect evidence to exclude bacterial infections. The primary outcome of the study was the response to SABA therapy, assessed by the improvement in SpO₂ and PEF measurements within 30 minutes to 1 hour following SABA administration. The SABA used in the study was Salbutamol (Ventolin®), manufactured by Glaxo-SmithKline. The resolution of the exacerbation was considered when the patient's SpO₂ levels returned to baseline and the PEF values improved by at least 20% from the pre-treatment measurement.

Factors influencing SABA effectiveness

Several factors were evaluated for their influence on the effectiveness of SABA during acute exacerbations, including BMI, age, gender, ICS adherence, and asthma exacerbation severity. BMI was classified according to the World Health Organization's criteria as underweight, normal weight, overweight, and obese. ICS adherence was measured using the Medication Adherence Report Scale (MARS-5), a questionnaire that was administered to the patients to assess their adherence to prescribed ICS therapy. Adherence was categorized as adherent (≥80% of prescribed doses) or non adherent (<80%). Exacerbation severity was assessed using the severity classification from the GINA, 2023, which divides exacerbations into mild, moderate, and severe categories based on the symptoms and the need for additional therapy beyond SABA (2).

SABA administration and other medications

The dose of salbutamol administered was 1 respule (2.5 mg) via nebulizer, with no other bronchodilators or treatments administered during the 1-hour observation period following SABA administration. Systemic glucocorticoids (e.g., methylprednisolone) were provided to some patients based on clinical need. The mean daily dosage of methylprednisolone was recorded, along with the number of days of therapy during the exacerbation period. Additionally, short-acting muscarinic antagonists (SAMAs), such as ipratropium

bromide, were given if necessary, and the specific doses and frequency of administration were documented. Oxygen therapy was administered to patients with SpO₂ values below 92%, and the flow rate and duration of oxygen administration were recorded. Adherence to ICS was assessed prior to the exacerbation, but no ICS or other therapies were administered during the observation period.

Statistical analysis

Descriptive statistics were used to summarize the demographic and clinical characteristics of the study population. The effectiveness of SABA therapy, measured by changes in SpO₂ and PEF, was compared across different subgroups defined by age, gender, BMI, and ICS adherence using paired t-tests for continuous variables and chi-square tests for categorical variables. A p-value of <0.05 was considered statistically significant. All statistical analyses were performed using SPSS version 25.0 (IBM Corporation, Armonk, NY).

Ethical considerations

This study was approved by the Ethics Committee of the Faculty of Medicine, Hasanuddin University, Makassar, under ethical clearance number 604/UN4.6.4.5.31/PP36/2025, and adhered to the principles of the Declaration of Helsinki. Informed consent was obtained from all participants prior to their inclusion in the study. All data were anonymized to ensure confidentiality and privacy.

Results

A total of 103 participants were enrolled in the study, with 57.3% females and 42.7% males. The majority of participants were non-elderly (74.8%), while 25.2% were elderly. The demographic characteristics of the study population, including age, body weight, height, BMI, nutritional status, and employment, are summarized in Table 1.

The mean BMI was within the normal range, with 73.8% of participants having normal nutritional status and 26.2% classified as obese. Regarding asthma

Table 1. Demographic Characteristics of the Study Population

Characteristic	n	%
Gender		
Male	44	42.7
Female	59	57.3
Age		
Mean $\pm$ SD	51 $\pm$ 10	
Non-Elderly (<60 years old)	77	74.8
Elderly ($\geq$ 60 years old)	26	25.2
IQ Range	58.1 – 44.6 = 13.5	
Body Weight (kg)		
Mean $\pm$ SD	61.6 $\pm$ 7.9	—
Min - Max	45.4 – 80.0	—
IQ Range	66.9 – 56.3 = 10.6 kg	
Height (cm)		
Mean $\pm$ SD	161.6 $\pm$ 7.6	—
Min - Max	145 – 185	—
BMI (kg/m ²)		
Mean $\pm$ SD	23.5 $\pm$ 2.3	—
Min - Max	19.2 – 29.6	—
IQ Range	66.9 - 56.3 = 10.6 kg	
Nutritional Status		
Non-Obese	76	73.8
Obese	27	26.2
Occupation		
Housewife	26	25.2
Self-Employed	28	27.2
Civil Servant	19	18.4
Retiree	9	8.7
Student	12	11.7
Others	9	8.7

severity, 62.1% of participants experienced mild to moderate exacerbations, while 37.9% had severe exacerbations. Adherence to ICS, as assessed by the Medication Adherence Report Scale (MARS-5), had an average score of 19.5 ± 3.4 , with 69.9% of participants adhering to prescribed ICS therapy. These

Table 2. Clinical Characteristics of the Study Population

Characteristic	n	%
ICS Adherence		
Adherent	72	69.9
Non-adherent	31	30.1
MARS-5 Score		
Mean $\pm$ SD	19.5 $\pm$ 3.4	—
Min - Max	12 – 25	—
Exacerbation Severity		
Mild-Moderate	64	62.1
Severe	39	37.9
Heart Rate (beats/min)		
Mean $\pm$ SD	84.3 $\pm$ 9.4	—
Min - Max	60 – 110	—
Respiratory Rate (breaths/min)		
Mean $\pm$ SD	26.0 $\pm$ 2.3	—
Min - Max	20 – 31	—
Body Temperature (°C)		
Mean $\pm$ SD	36.7 $\pm$ 0.3	—
Min - Max	36.0 – 37.8	—
Respiratory Infection		
Present	29	28.2
Absent	74	71.8
Exacerbation Resolution Status		
Resolved	66	64.0
Not Resolved	37	36.0

clinical characteristics, including medication adherence, asthma exacerbation severity, and vital signs, such as heart rate, respiratory rate, and body temperature, along with the presence of concurrent respiratory infections (28.2%), are detailed in Table 2.

Following treatment, 68.0% of participants achieved complete resolution of their exacerbations, while 32.0% had unresolved symptoms, suggesting a positive treatment outcome for the majority of the study population. The results of the paired tests for oxygen saturation (SpO₂) and PEF before and after SABA administration are summarized in Table 3.

Table 3. Oxygen Saturation and Peak Expiratory Flow (PEF) Before and After SABA Administration

	Before	After	Change (Δ)	p-value
SpO ₂ (%)	91.2 $\pm$ 4.0	97.0 $\pm$ 2.3	5.8 $\pm$ 4.7	< 0.001 ^{a*}
PEF (L/min)	250.7 $\pm$ 59.2	331.6 $\pm$ 77.1	80.9 $\pm$ 48.9	< 0.001 ^{b*}
PEF (% predicted)	52.8 $\pm$ 12.8	70.1 $\pm$ 16.4	17.3 $\pm$ 10.6	< 0.001 ^{b*}

Statistically significant: a Wilcoxon test, b Paired t-test

Statistically significant: a Wilcoxon test, b Paired t-test

Significant improvements were observed in both SpO₂ and APE following SABA treatment. The average SpO₂ increased from 91.2% $\pm$ 4.0 to 97.0% $\pm$ 2.3, reflecting a mean improvement of 5.8%. Similarly, the APE (in L/min) showed a substantial rise from 250.7 $\pm$ 59.2 L/min to 331.6 $\pm$ 77.1 L/min, corresponding to an increase of 80.9 L/min. When expressed as a percentage of the predicted value, APE improved from 52.8% $\pm$ 12.8 to 70.1% $\pm$ 16.4, demonstrating a notable enhancement in lung function. All of these improvements were statistically significant, with p-values < 0.001, confirming the effectiveness of SABA in addressing acute asthma exacerbations by significantly improving oxygenation and relieving airway obstruction. Bivariate analysis using the Chi-Square test revealed several factors significantly associated with the successful resolution of asthma exacerbations. ICS adherence was a strong protective factor, with adherent patients having 5.9 times higher odds of resolution compared to non-adherent patients (p < 0.001). In contrast, severe exacerbation (OR: 0.286; p = 0.003), respiratory infections (OR: 0.105; p < 0.001), and obesity (OR: 0.263; p = 0.003) were identified as risk factors reducing the chances of resolution. Demographic variables, such as age (elderly vs. non-elderly) and gender, were not significantly associated with resolution outcomes, as shown in Table 4.

Analysis of Δ SpO₂ following intervention revealed significant differences across various patient characteristics. Greater improvements in SpO₂ were observed in ICS adherent patients, those with severe exacerbations, individuals with respiratory infections, those with obesity, and male patients. Interestingly, despite poorer resolution outcomes in these groups (Table 5), they exhibited greater potential for

improvement in oxygen saturation. Age did not show a significant difference in Δ SpO₂ between elderly and non-elderly groups (Table 5).

Further comparison of Δ PEF, both in L/min and as a percentage of predicted values, showed that ICS adherence remained a critical factor, with adherent patients demonstrating significantly greater improvements in PEF than non-adherent patients. Nutritional status also played a role, with non-obese patients showing better improvement in lung function (Δ PEF) than their obese counterparts. Although the change in PEF (L/min) was not significantly different in patients with respiratory infections, the improvement in predicted PEF was significantly lower in the infection group. Exacerbation severity and gender did not show significant differences in PEF improvement, indicating that the response to SABA therapy in terms of expiratory flow was relatively consistent across severity levels and genders, as indicated in Table 6.

Multivariate logistic regression revealed that ICS adherence was a significant predictor of asthma exacerbation resolution. Patients adhering to ICS therapy had 5.3 times higher odds of resolution compared to non-adherent patients (adjusted OR: 5.289; 95% CI: 1.797–15.565; p = 0.002). In contrast, respiratory infections were a strong risk factor for poor resolution, with patients experiencing respiratory infections having 8.6 times lower odds of resolution (adjusted OR: 8.613; 95% CI: 2.848–26.055; p < 0.001). While asthma exacerbation severity and nutritional status (obesity) were significant in the bivariate analysis, these factors lost significance in the multivariate model. Specifically, asthma severity did not remain a significant predictor of exacerbation resolution (p = 0.167), and obesity did not influence the outcome after adjusting for ICS adherence and respiratory infections. These findings are summarized in Table 7.

Table 4. Clinical Factors and Asthma Exacerbation Resolution Status

Variable	Exacerbation Resolution Status		OR (95% CI)	p-value
	Resolved n (%)	Not Resolved n (%)		
ICS Adherence				
Non-adherent	11(35.5)	20(64.5)	5.882(2.356-14.686)	< 0.001*
Adherent	55(76.4)	17(23.6)		
Asthma Exacerbation Severity				
Severe	18(46.2)	21(53.8)	3.500 (1.501-8.160)	0.003*
Mild-Moderate	48(75.0)	16(25.0)		
Respiratory Infection				
Present	8(27.6)	21(72.4)	9.516 (3.555-25.471)	< 0.001*
Absent	58(78.4)	16(21.6)		
Nutritional Status				
Obese	11(40.7)	16(59.3)	3.810 (1.522-9.537)	0.003*
Non-Obese	55(72.4)	21(27.6)		
Age Category				
Elderly	18(69.2)	8(30.8)	0.736 (0.284-1.906)	0.526
Non-Elderly	48(62.3)	29(37.7)		
Gender				
Female	42(71.2)	17(28.8)	0.486 (0.214-1.101)	0.082
Male	24(54.5)	20(45.5)		

*Chi-Square test, Nagelkerke R² = 0.445, *Significant*

Table 5. Mann-Whitney U Test Comparing ΔSpO_2 Based on Factor Variables

Variable	Category	Median ΔSpO_2 (%) (Min.-Max.)	p-value
ICS Adherence	Adherent	5.0 (-3.0 – 19.0)	0.026*
	Non-adherent	3.0 (-4.0 – 16.0)	
Asthma Exacerbation Severity	Severe	13.0 (-4.0 – 19.0)	< 0.001*
	Mild-Moderate	4.0 (-3.0 – 8.0)	
Respiratory Infection	Present	12.0 (0 – 19.0)	0.017*
	Absent	4.0 (-4.0 – 16.0)	
Nutritional Status	Obese	13.0 (-4.0 – 19.0)	0.011*
	Non-obese	4.0 (-3.0 0 15.0)	
Age Category	Elderly	5.0 (-3.0 – 16.0)	0.377
	Non-Elderly	4.0 (-4.0 – 19.0)	
Gender	Male	6.0 (-4.0 – 19.0)	< 0.001*
	Female	4.0 (-3.0 – 15.0)	

*Mann-Whitney U test, *Significant*

Table 6. Comparing Δ PEF Based on Variables

Variable	Median Δ PEF (L/min) (Min.-Max.)	p-value	Median Δ PEF (%) (Min.-Max.)	p-value
ICS Adherence				
Adherent	86.0 (6.0 – 242.0)	< 0.001*	18.2 (1.4 – 60.8)	0.003*
Non-adherent	52.0 (-8.0 – 156.0)		12.8 (0.8 – 31.7)	
Asthma Exacerbation Severity				
Severe	71.0 (18.0 – 242.0)	0.865	15.3 (1.4 – 44.8)	0.523
Mild-Moderate	76.5 (-8.0 – 218.0)		17.6 (0.8 – 60.9)	
Respiratory Infection				
Present	57.0 (5.0 – 187.0)	0.061	11.7 (2.6 – 39.8)	0.044*
Absent	82.5 (-8.0 – 242.0)		17.6 (0.8 – 60.8)	
Nutritional Status				
Obese	59.0 (6.0 – 218.0)	0.037*	11.7 (1.6 – 60.8)	0.009*
Non-obese	81.0 (-8.0 – 242.0)		17.9 (0.8 – 44.8)	
Age Category				
Elderly	80.5 (-8.0 – 178.0)	0.511	15.9 (0.8 – 39.8)	0.699
Non-elderly	72.0 (5.0 – 242.0)		15.6 (1.4 – 60.8)	
Gender				
Male	80.5 (15.0 – 242.0)	0.357	13.7 (2.8 – 44.8)	0.490
Female	70.0 (-8.0 – 218.0)		17.4 (0.8 – 60.9)	

*Mann-Whitney U test, *Significant***Table 7.** Multivariate Analysis of Factors Influencing Asthma Exacerbation Resolution Status

Variable	Asthma Exacerbation Resolution Status		p-value	Adjusted OR (95% CI)
	Resolved n(%)	Not Resolved n(%)		
ICS Adherence				
Adherent	11(35.5)	20(64.5)	0.002*	5.289 (1.797-15.565)
Non-adherent	55(76.4)	17(23.6)		
Asthma Exacerbation Severity				
Severe	18(46.2)	21(53.8)	0.167	-
Mild-Moderate	48(75.0)	16(25.0)		
Respiratory Infection				
Present	8(27.6)	21(72.4)	< 0.001*	8.613 (2.848-26.055)
Absent	58(78.4)	16(21.6)		
Nutritional Status				
Obese	11(40.7)	16(59.3)		
Non-Obese	55(72.4)	21(27.6)		

*Logistic Regression, *Significant

Discussion

The results of this study provide strong evidence for the effectiveness of SABA in managing acute asthma exacerbations. The administration of SABA resulted in significant improvements in critical clinical markers, particularly oxygen saturation (SpO_2) and PEF. Specifically, SpO_2 increased from $91.2 \pm 4.0\%$ to $97.0 \pm 2.3\%$, and PEF improved from 250.7 ± 59.2 L/min to 331.6 ± 77.1 L/min. This is consistent with the established role of SABA in acute exacerbations, as highlighted by international guidelines such as those from the GINA, which advocate for SABA as the cornerstone treatment due to their rapid onset of action and ability to quickly relieve bronchoconstriction (2). Our findings align with previous studies that have demonstrated similar improvements in respiratory function following SABA administration. For instance, Andrade et al. (2007) found significant increases in both SpO_2 and PEF following salbutamol treatment in pediatric patients. Specifically, their study showed an improvement in SpO_2 from $93.6 \pm 2.9\%$ to $95.0 \pm 2.7\%$ and an increase in predicted PEF from $61.8 \pm 10.6\%$ to $80.2 \pm 13.1\%$ (13). Though there are differences in population characteristics and exacerbation severity between their study and ours, the pattern of improvement in both SpO_2 and PEF observed in both studies highlights the therapeutic value of SABA in acute settings. Additionally, a study by Onubogu et al. (2022) confirmed the utility of SpO_2 and PEF as reliable indicators for monitoring the response to SABA therapy. They reported a moderate correlation between SpO_2 and PEF ($r = 0.44$, $p = 0.04$) during asthma exacerbations, further supporting our use of these parameters to evaluate the immediate effects of SABA on airway function (14).

The crucial role of ICS adherence in acute asthma management

In our study, adherence to ICS was found to be a pivotal factor influencing the effectiveness of SABA therapy. Patients who were adherent to their ICS regimen demonstrated significantly better improvements in both PEF and SpO_2 compared to non-adherent patients. Specifically, ICS-adherent patients exhibited a mean increase of 90.50 ± 48.65 L/min in PEF,

compared to 58.64 ± 42.50 L/min in non-adherent patients ($p = 0.001$). Furthermore, the increase in predicted PEF was also greater in the adherent group ($19.28 \pm 10.91\%$) compared to the non-adherent group ($12.54 \pm 8.19\%$; $p = 0.003$).

These findings are consistent with the growing body of literature emphasizing the importance of ICS adherence in enhancing bronchodilator responses. A meta-analysis by Chan et al. (2023) demonstrated that adherence to ICS/LABA therapy significantly improved medication adherence by 23%, and it was associated with a 47% reduction in the risk of severe asthma exacerbations (15). This relationship between ICS adherence and improved clinical outcomes is further supported by the systematic review conducted by Jeminiwa et al. (2019), which showed that adherence to ICS is positively correlated with better bronchodilator responsiveness, particularly during acute asthma episodes (16). The mechanisms by which ICS enhance the efficacy of SABA are well understood. ICS exert their effects by reducing inflammation in the airways, primarily through the inhibition of pro-inflammatory transcription factors like NF- κ B and AP-1. This reduces the production of cytokines such as IL-4, IL-5, and IL-13, as well as the infiltration of inflammatory cells like eosinophils and T-cells. This suppression of chronic inflammation creates a more responsive airway, which is crucial for optimizing the effects of bronchodilators (17). ICS also help prevent the downregulation of β_2 -adrenergic receptors, which is common in patients with persistent airway inflammation. By maintaining receptor sensitivity, ICS ensure that SABA can exert its maximum bronchodilatory effect (18). Given these mechanisms, it is evident that improving ICS adherence not only helps in controlling chronic symptoms but also maximizes the effectiveness of rescue medications like SABA. Strategies to improve adherence, such as the use of electronic monitoring devices (EMDs) for real-time feedback, have shown promise in increasing adherence by up to 23% (19).

Respiratory infections as a key modulator of asthma exacerbation resolution

The presence of respiratory infections significantly impacted the resolution of asthma exacerbations in our

study. Patients with concurrent respiratory infections had substantially lower odds of resolution (adjusted OR: 0.116; 95% CI: 0.038–0.351; $p < 0.001$). This finding is consistent with existing literature, which underscores the detrimental effect of respiratory infections on asthma outcomes. Latorre et al. (2023) also highlighted the exacerbating role of respiratory infections, showing that they impede the resolution of exacerbations and complicate asthma management (20). Respiratory infections exacerbate airway inflammation, leading to increased mucus production and bronchoconstriction. These changes create an environment where the airways become less responsive to bronchodilators, thus limiting the effectiveness of SABA (17). Interestingly, while patients with infections had poorer resolution outcomes overall, they demonstrated a larger potential for improvement in SpO₂, suggesting that they still responded to bronchodilator therapy, albeit less effectively. This finding emphasizes the importance of addressing respiratory infections early in exacerbations to enhance the efficacy of SABA therapy. Overall, these results highlight the complex interaction between medication adherence, infection control, and asthma exacerbation resolution. The ability of ICS to enhance the therapeutic effects of SABA, particularly in the presence of respiratory infections, underscores the importance of comprehensive asthma management strategies that address both chronic inflammation and infection prevention.

The effect of asthma exacerbation severity on short-acting β_2 -agonist response

The findings from our study indicate that the severity of asthma exacerbations significantly influences the effectiveness of SABA. Patients experiencing severe exacerbations showed a significantly greater improvement in oxygen saturation (SpO₂) after SABA treatment compared to those with mild to moderate exacerbations. Specifically, the average SpO₂ increase for severe exacerbations was $9.05 \pm 5.83\%$, compared to only $3.76 \pm 2.03\%$ for mild to moderate cases ($p = 0.001$). This observation is consistent with the “ceiling effect,” a concept in which patients with more severe baseline hypoxia (lower SpO₂) have more room for improvement following bronchodilation, even though

their final SpO₂ remains lower than that of patients with less severe exacerbations. This effect suggests that individuals with more pronounced hypoxia before treatment have a greater capacity for improvement in oxygen levels once the airway obstruction is relieved. However, despite the improvements in SpO₂, no significant differences in PEF were observed between the groups, both in liters per minute ($p = 0.865$) and in percentage of predicted values ($p = 0.523$). This finding further supports the distinct roles that SpO₂ and PEF play in evaluating asthma exacerbations. While SpO₂ reflects overall oxygenation, making it more sensitive to changes in exacerbation severity, PEF primarily measures the function of larger airways and is less responsive to acute shifts in oxygenation levels. Previous studies have similarly highlighted that SpO₂ can be more influenced by changes in both small and large airways, whereas PEF specifically reflects the condition of the larger, proximal airways. In the multivariate analysis, the significance of exacerbation severity as a predictor of asthma resolution outcomes diminished after adjusting for ICS adherence and the presence of respiratory infections. This suggests that the impact of exacerbation severity may be mediated by other factors, such as medication adherence and comorbid respiratory infections, rather than being an independent predictor of resolution. These findings align with the work of Denlinger et al. (2017), who demonstrated that factors like gastroesophageal reflux disease (GERD), obesity, and eosinophil counts are more predictive of asthma exacerbation outcomes than baseline severity (21). This underscores the importance of managing comorbidities and ensuring proper medication adherence, particularly during severe exacerbations, to improve patient outcomes.

The role of obesity in modulating SABA effectiveness during asthma exacerbations

In our cohort, 26.2% of participants were classified as obese (BMI ≥ 25 kg/m²). Obese patients showed a significantly larger increase in SpO₂ ($8.63 \pm 6.31\%$) compared to non-obese patients ($4.75 \pm 3.46\%$; $p = 0.011$). However, their improvement in PEF was less pronounced, both in liters per minute (67.81 ± 52.14 L/min vs. 85.56 ± 47.23 L/min; $p =$

0.037) and in percentage of predicted PEF ($13.91 \pm 12.09\%$ vs. $18.44 \pm 9.82\%$; $p = 0.009$). These findings indicate that while obesity is associated with improvements in oxygenation, it limits the improvement in airway function, as reflected by the smaller increase in PEF. Obesity is well-established as an independent risk factor for worsened asthma control, more frequent exacerbations, and poor response to treatment. Studies have shown that obesity increases the likelihood of severe asthma exacerbations by 1.5 to 1.9 times, with a clear dose-response relationship (22). The mechanisms through which obesity impairs asthma control are complex. Obesity leads to systemic inflammation, characterized by elevated levels of adipokines such as leptin and reduced adiponectin, as well as increased pro-inflammatory cytokines like IL-6 and TNF- α . This inflammatory response promotes a shift toward a neutrophilic and Th17-driven immune profile, which is less responsive to conventional therapies such as ICS and SABA (23). Obesity also contributes to metabolic dysfunctions, including insulin resistance and hyperglycemia, which in turn cause airway smooth muscle proliferation, epithelial damage, and worsened airway hyperresponsiveness (AHR). These factors contribute to the reduction in SABA effectiveness. Furthermore, obesity leads to mechanical changes in the lungs, such as reduced lung compliance and decreased expiratory reserve volume, due to the compressive effects of excess adipose tissue. These changes result in increased collapsibility of peripheral airways, making it more difficult to relieve airway obstruction during asthma exacerbations (24). Although obesity was not an independent predictor of exacerbation resolution after adjusting for ICS adherence and respiratory infection status in our multivariate analysis, it remains an important factor to consider in asthma management. Our findings suggest that the negative impact of obesity on asthma outcomes may be mediated by factors such as poor medication adherence and increased susceptibility to infections. These findings align with the Severe Asthma Research Program (SARP), where obesity was found to be significant in bivariate analysis but lost significance in the multivariate model after adjusting for eosinophilia, GERD, and adherence to therapy (25).

How respiratory infections influence the efficacy of SABA in asthma control

Respiratory infections had a profound effect on asthma exacerbation outcomes in this study. Of the 103 participants, 28.2% had concurrent respiratory infections during their exacerbations. These patients exhibited greater improvements in SpO₂ ($8.72 \pm 5.84\%$) compared to those without infections ($4.61 \pm 3.56\%$; $p = 0.017$). However, the improvement in PEF was less substantial in the infection group ($14.09 \pm 9.46\%$ vs. $18.49 \pm 10.81\%$; $p = 0.044$). While respiratory infections led to more severe baseline hypoxia, they also limited the improvement in airway function, suggesting a complex interaction between infection and airway function. Respiratory infections, especially viral ones like rhinovirus, exacerbate asthma by increasing airway inflammation, impairing mucociliary clearance, and amplifying airway hyperresponsiveness (26). These infections lead to increased inflammation in the airways, which reduces the effectiveness of bronchodilators like SABA. This is consistent with our findings, where respiratory infections were strongly associated with reduced odds of asthma exacerbation resolution, with an adjusted OR of 0.116 (95% CI: 0.038–0.351; $p = 0.001$). Although patients with infections demonstrated greater potential for improvement in SpO₂, this group exhibited limited improvements in PEF. This suggests that while infections exacerbate asthma and complicate long-term control, they allow for immediate improvements in oxygenation when treated with effective bronchodilators. This highlights the need for a comprehensive approach to asthma exacerbation management, where both infection control and bronchodilator therapy are critical to optimizing patient outcomes.

Strengths and limitations

This study provides important insights into the factors influencing the effectiveness of SABA in managing acute asthma exacerbations. By evaluating key clinical parameters such as ICS adherence, asthma severity, and respiratory infections, it offers a comprehensive understanding of the determinants of SABA response. The real-time data on SpO₂ and PEF improvements, along with multivariate analysis, strengthen the reliability of the

findings by accounting for potential confounders. However, the study has several limitations. The sample size of 103 participants limits statistical power, particularly within subgroups like the elderly or obese. The cross-sectional design restricts the ability to draw causal conclusions, especially regarding ICS adherence and respiratory infections. The absence of long-term follow-up prevents the assessment of recurrent exacerbations or sustained asthma control. Additionally, the cohort's demographic, primarily younger adults with mild to moderate exacerbations, may limit the generalizability of the results. The lack of inflammatory biomarkers, such as eosinophils or FeNO, further limits the ability to explore asthma phenotypes and the role of inflammation in treatment response. Although respiratory infections were clinically diagnosed, the absence of laboratory confirmation introduces potential misclassification bias. ICS adherence was assessed using the MARS-5 questionnaire, which relies on self-reporting. While this method provides useful insights into perceived adherence, it is subjective and may not fully reflect actual medication use. More objective adherence measures, such as electronic monitoring, would have offered a more accurate assessment, but these tools were unavailable for this study. Therefore, the adherence data should be interpreted with caution. To address these limitations, future research should involve larger, more diverse populations and incorporate inflammatory biomarkers to enhance understanding of SABA effectiveness in asthma exacerbations.

Conclusion

In conclusion, this study underscores the critical role of short-acting β_2 -agonists (SABA) in managing acute asthma exacerbations, with significant improvements observed in key clinical markers such as oxygen saturation (SpO_2) and peak expiratory flow (PEF) following SABA administration. ICS adherence emerged as a key determinant of SABA effectiveness, with adherent patients exhibiting notably better responses. In contrast, the presence of respiratory infections significantly impaired the resolution of exacerbations, highlighting the need for comprehensive management that addresses both medication adherence and infection control to optimize treatment outcomes. While

exacerbation severity and obesity initially appeared to influence outcomes, these factors lost significance in the multivariate analysis after accounting for ICS adherence and respiratory infection status. This suggests that, beyond exacerbation severity, ICS adherence and infection prevention are paramount in improving asthma resolution. Thus, effective management of acute asthma exacerbations should focus on optimizing ICS adherence and controlling respiratory infections to ensure better clinical responses, particularly in patients with more severe exacerbations or comorbid conditions.

Ethic approval: Ethical approval was obtained from the Ethics Committee of the Faculty of Medicine, Hasanuddin University, Makassar, under ethical clearance number 604/UN4.6.4.5.31/PP36/2025.

Conflict of interest: Each author declares that he or she has no commercial associations (e.g. consultancies, stock ownership, equity interest, patent/licensing arrangement etc.) that might pose a conflict of interest in connection with the submitted article.

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